

CLASS :
EXAM NO :

TIME : 3 Hrs
MARKS : 80

SECTION A (1 mark each)

(16 × 1 = 16)

1) What structure helps ctenophores to swim?

- a) Tentacles
- b) Flagella
- c) Ciliated comb plates
- d) Muscular foot

2) Which of the following is a characteristic feature of ctenophores?

- a) They possess a coelom
- b) They are bilaterally symmetrical
- c) They exhibit bioluminescence
- d) They have a true skeleton

3) Which organ of the frog functions as the conduit for both urinary and reproductive wastes?

- a) Kidney
- b) Cloaca
- c) Lymphatic vessels
- d) Brain

4) Which of the following is NOT a characteristic feature of Ctenophora?

- a) Radial symmetry
- b) Presence of comb plates
- c) Bilateral symmetry
- d) Bioluminescence

5) How do ctenophores move through the water?

- a) By jet propulsion
- b) By ciliated comb plates
- c) By muscular contractions
- d) By walking on substrate

6) Which of the following is a characteristic feature of ctenophores?

- a) Radial symmetry
- b) Bilateral symmetry
- c) Segmentation
- d) Notochord

7) Ctenophores perform digestion that is:

- a) Only extracellular
- b) Only intracellular
- c) Both extracellular and intracellular
- d) None of the above

8) The organ through which frogs excrete both urine and waste material from the digestive tract is known as the

- a) cloaca
- b) vent
- c) rectum
- d) anus

9) Which structure in ctenophores aids in their locomotion?

- a) Comb plates
- b) Tentacles
- c) Cilia on the body surface
- d) Bioluminescent organs

10) Which of the following statements about ctenophores is correct?

- a) They possess a coelom.
- b) They are diploblastic.
- c) They have a true nervous system.
- d) They are terrestrial.

11) The cloacal aperture in frogs is located at the ventral end of which body region?

- a) thorax
- b) abdomen
- c) pelvis
- d) head

12) Which of the following tissues is responsible for storing starch in a plant?

- a) Phloem
- b) Xylem
- c) Starch sheath
- d) Cortex

13) The movement of water in the xylem of a plant is mainly due to

- a) Capillary action
- b) Cohesion–tension mechanism
- c) Osmotic pressure
- d) Diffusion

14) In the cell cycle, the phase in which DNA is replicated is called

- a) G₀
- b) G₁
- c) S
- d) M

15) The outer layer of the fruit, which is usually green and later turns yellow, is known as

- a) Endocarp
- b) Exocarp
- c) Mesocarp
- d) Pericarp

16) Which of the following is not a component of the blood in humans?

- a) Plasma
- b) Platelets
- c) Lymphocytes
- d) Red blood cells

SECTION B (2 marks each)

(5 × 2 = 10)

**17) Which of the following is NOT a respiratory organ in a frog? A. Skin B. Lungs
C. Buccal cavity D. Gall bladder**

**18) Which of the following is not a component of the frog's excretory system? A.
Kidneys B. Ureters C. Urogenital duct D. Lymph nodes**

**19) Which of the following best describes the function of the chloroplast in plant
cells?**

SECTION C (3 marks each)

(7 × 3 = 21)

**22) Identify the function of one of the following endocrine glands in a frog: (a)
thyroid, (b) adrenal.**

**23) Which of the following statements about the frog's reproductive system is
correct? a) The male frog's testes are situated in the peritoneal cavity. b) The
female frog has a pair of ovaries that produce eggs. c) The frog's cloaca functions
as a combined excretory, urinary, and reproductive opening. d) Both a and b are
correct**

**24) Describe the role of chloroplasts in the process of photosynthesis. Explain
why the presence of light is crucial for this process, and suggest two alternative
energy sources that could potentially be used by chloroplasts.**

25) The cloaca in frogs serves which of the following functions?

- a) Passage of urine and fecal matter only
- b) Passage of urine, fecal matter, and reproductive material
- c) Passage of fecal matter and reproductive material only
- d) Passage of urine, fecal matter, reproductive material, and egg storage

SECTION D (4 marks each)

(2 × 4 = 8)

**29) Ctenophores, commonly known as comb jellies, are exclusively marine
animals that display radial symmetry and possess a diploblastic body plan. Their
bodies are composed of two germ layers – ectoderm and endoderm – with a
gelatinous mesoglea in between. Eight external rows of ciliated comb**

SECTION E (5 marks each)

(3 × 5 = 15)

**31) Which of the following structures is a part of the frog's respiratory system that
functions primarily in aquatic respiration? a) Cutaneous skin b) Lung c) Buccal
cavity d) Both a and c**

32) Describe the key structural and functional features of ctenophores that enable them to thrive in marine environments, including their body symmetry, level of tissue organisation, and the role of comb plates in locomotion.

33) Ctenophores exhibit both extracellular and intracellular digestion. Choose one of the following and explain in detail:

(A) The process and advantages of extracellular digestion in ctenophores, or

(B) The mechanism of intracellular digestion and

ANSWER KEY (SECTION A)

- Q1. (c)
- Q2. (c)
- Q3. (b)
- Q4. (c)
- Q5. (b)
- Q6. (a)
- Q7. (c)
- Q8. (a)
- Q9. (a)
- Q10. (b)
- Q11. (c)
- Q12. (c)
- Q13. (b)
- Q14. (c)
- Q15. (b)
- Q16. (c)